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Breast cancer trends differ by ethnicity: a report from the South African National Cancer Registry (1994–2009)

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Background: To describe breast cancer (BC) incidence and mortality by ethnicity in South Africa (SA). **Methods:** Sources of data included the South African National Cancer Registry (NCR) pathology-based reports (1994–2009) and Statistics South Africa (SSA) mortality data (1997–2009). Numbers of cases, age-standardised incidence rates (ASIR) and lifetime risk (LR) were extracted from the NCR database for 1994–2009. Age-specific incidence rates were calculated for five-year age categories. The direct method of standardisation was employed to calculate age-standardised mortality rates (ASMR) using mortality data. **Results:** Between 1994 and 2009, there were 85 561 female BC. For the Black, Coloured and Asian groups, increases in ASIR and LR were observed between 1994 and 2009. In 2009, the ASIR for the total population, Blacks, Whites, Coloureds and Asians were 26.9, 18.7, 50.2, 40.9 and 51.2 per 100 000, respectively. For Asians, an increase in proportion of BC as a percentage of all female cancers was observed between 1994 and 2002 (11.1%) and continued to increase to 2009 (a further 4.5%). Whites and Asians presented higher incidences of BC at earlier ages compared with Blacks and Coloureds in 2009. In 1998, there were 1618 BC deaths in SA compared with 2784 deaths in 2009. ASMR between 1997 and 2004 increased but stabilised thereafter. **Conclusion:** This paper demonstrated that SA BC incidence rates are similar to other countries in the region, but lower than other countries with similar health systems. Ethnic differences in BC trends were observed. However, the reasons for observed ethnic differences are unclear.

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Introduction

Breast cancer (BC) is one of the commonest female cancers in women worldwide. Globally, an estimated 1.67 million incident cases were diagnosed in 2012 corresponding to a rate of 43/100 000^{1–3} with 6.3 million prevalent BC in the previous 5 years.^{1,4} Between 1980 and 2010, the annual BC incidence increase worldwide was 3.1%.⁵ Projections for 2030 predict a significant shift in global cancer distribution towards low- and middle-income countries (LMIC), with increases occurring in all age groups, as a result of the population growth, increased life expectancy and adoption of westernised lifestyles.^{1,4,6–9}

GLOBOCAN estimated that 522 000 females (13/100 000 population) died from BC during 2012, ranking BC as the fifth leading cause of cancer death.¹ Sixty-two percent of BC deaths were estimated to have occurred in LMIC's, in large part due to late presentation/diagnosis and inadequate access to treatment.^{1,10} Monitoring BC burden is important to successfully plan healthcare interventions. An important source of population-level cancer data originates from cancer registries. Despite the fact that LMIC bear an increasing cancer burden, less than 10% of people in LMIC are covered by high-quality population-based cancer registries.⁷ African data are scarce with trend analyses usually covering the USA, European Union, Asia and Latin America.⁵ Only four population-based cancer registries (PBCR) in sub-Saharan Africa (Zimbabwe, Uganda, Malawi and Eastern Cape in South Africa) were of sufficient quality to be published within the International Agency of Research on Cancer's (IARC) monograph 'Cancer Incidence in Five Continents Volume X'.¹¹

South Africa (SA) is a middle-income country (MIC) with a dual health care system (private and public healthcare facilities) and multi-racial demographics. Health status is influenced by many factors including inequities brought about by the apartheid past and the socio-economic status of individuals which is largely divided along racial lines.¹²

Although the South African National Cancer Registry (NCR) has been in operation since 1986 and is the only source of national population-level cancer data for the country, very little has been reported on SA's national cancer burden. BC is the leading cancer amongst women in SA, and the second commonest cancer amongst Black women, after cervical cancer.¹³ A previous description of BC in SA women for 1993–95 reported an age-standardised incidence rate (ASIR) of 25/100 000 for all population groups with a lifetime risk (LR) of 1 in 36.¹⁴ This paper reports on more recent SA data on BC, explores BC trends (incidence and mortality rates) and investigates differences by ethnicity in post-apartheid SA between 1994 and 2009.

Methods

Sources of data included the NCR reports (1994–2009) and mortality data from SSA for 1997–2009.

NCR methodology

The NCR is a pathology-based cancer registry that collects all cytologically and histologically diagnosed invasive malignancies from private and public healthcare laboratories in SA.¹⁵ For all years, pathology reports were received in electronic or hard copy format, and demographic and tumour information were extracted. No information on tumour stage or grade was captured, as these variables were incomplete from the pathology reports received. Where population group was missing, a hot-deck imputation method was used to impute data.¹⁶ Topographies and morphologies were coded according to the International Classification of Diseases for Oncology Version 3 (ICD-O3). Duplicate cases were removed by reviewing the entire database since inception in order to retain

only incident cases of cancer. Quality control measures included checks for improbable codes and duplicate coding of a percentage of cases. Death certificate data were not included as the NCR does not have access to confidential vital registration data. The methodology of the NCR has been comprehensively described previously by Singh *et al.*¹⁷

Statistical analysis

The NCR reports crude and direct ASIR for all cancers, 95% confidence intervals and cumulative LR (likelihood of developing cancer in one's lifetime, if one lives to age 74 years), by gender and population group.¹⁸

For this analysis, frequency, ASIR (per 100 000 persons) and LR (1:x number of cases) per population group were extracted from the NCR database for 1994–2009 (www.ncr.ac.za). The population groups included Black, White, Coloured (mixed race) and Asian/Indian, as used for the SA Census 2011.¹⁹ The terms 'population group', 'race group' and 'ethnic group' are used interchangeably in this paper to refer these four classifications of the SA population. ASIR for White and Coloured women were excluded for 1996/97 as the NCR aggregated figures for these groups for those years. Age-specific incidence rates were calculated using mid-year population estimates (from the Centre for Actuarial Research) as denominators for 5-year age categories.²⁰ The direct method of standardisation (Segi World Standard Population) was employed to calculate age-standardised mortality rates (ASMR) using data provided by SSA for 1997–2009 and mid-year population estimates from the Centre for Actuarial Research.²⁰ Mortality is reported for all females but stratification by ethnicity was not possible as this information was not available from SSA.

Mortality incidence ratios (MIR) were calculated as ASMR divided by ASIR to estimate a standard, population-based measure of fatality.²¹

Ethics

The NCR was granted an ethics waiver (Number: W-AW_220909-2) by the University of Witwatersrand.

Results

Incidence

Between 1994 and 2009, 85 561 female BC cases were reported to the NCR. The highest number of cases reported for the 15-year period was amongst the Black population ($n = 33,956$), followed by Whites ($n = 33,011$), Coloureds ($n = 12,538$) and Asians ($n = 4,231$). In comparison in 2011, the distribution of the population groups in SA was 79.2% Black, 8.9% Whites, 8.9% Coloured and 2.5% Asians. This distribution has been relatively stable since 1996.²² Figure 1

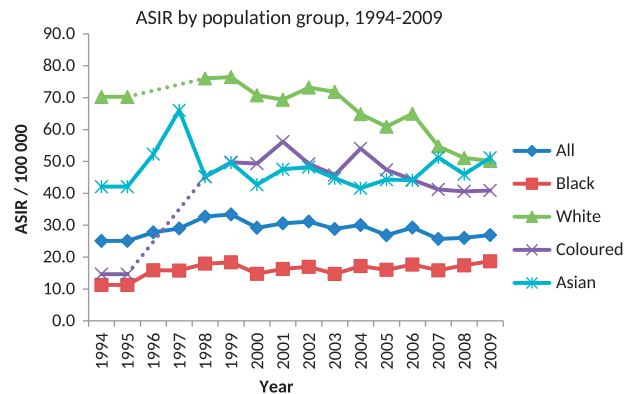


Figure 1 ASIR by population group, 1994–2004 (dotted lines indicate the missing data due to aggregation of White and Coloured groups for NCR reporting in 1996–97)

illustrates ASIR for female BC by ethnicity between 1994 and 2009. In 2009, the ASIR for the total population, Blacks, Whites, Coloureds and Asians were 26.9, 18.7, 50.2, 40.9 and 51.2 per 100 000, respectively. For all females, ASIR's between 1994 and 2009 were comparable (25.1 vs. 26.9 per 100 000). For the Black, Coloured and Asian groups, increases in ASIR were observed from 11.3 to 18.7 per 100 000 in Blacks, 14.7 to 40.9 in Coloureds and 42.1 to 51.2 in Asians between 1994 and 2009, respectively. For White females, a decrease in ASIR was observed from 70.2 to 50.2 per 100 000 between 1994 and 2009. From 1998 onwards, there appears to be an overall stabilisation of BC for the Black, Coloured and Asian populations.

LR for BC (all cases) increased from 1:36 in 1994 to 1:33 in 2009 (Supplementary table S1). The Black and Coloured population groups demonstrated the biggest increase in LR between 1994 and 2009, from 1:81 to 1:49 for Blacks and 1:63 to 1:22 for Coloured populations, respectively.

Table 1 illustrates BC as a proportion of all female cancers (including basal cell and squamous cell carcinoma of the skin) between 1994 and 2009 by ethnicity. For all females, an increase in percentage of BC proportion was observed between 1994 and 2002 (from 17.3 to 20.8%) but stabilised through to 2009. Black and Coloured population groups showed an increase of 5.4 and 4.0%,

respectively, between 1994 and 2002, then both stabilised through to 2009, whilst the White group remained stable from 1994 to 2009. For Asians, an increase in proportion of BC was observed between 1994 and 2002 (11.1%), and continued to increase to 2009 by another 4.5%. Age-specific incidence rates for 2009 are reported in figure 2. In this study, age-specific incidence illustrates the age range where incidence rates for specific ethnic groups become of concern. If an incident rate of 40 per 100 000 were regarded as a threshold of concern, we observe that Blacks, Whites, Coloureds and Asians had this rate (40/100 000) at the following age ranges: 45–49, 35–39, 40–44 and 35–39 years, respectively (figure 2). This illustrates that Whites and Asians presented higher incidences rates of BC at an earlier ages as compared with Blacks and Coloureds in 2009. Between the ages of 45–84 years, an increase of about 10, 50, 121 and 180 per 100 000 was observed for Blacks, Whites, Coloured and Asians, respectively, in 2009. Comparisons of age-specific incidence rates per population group for three time points (1994, 2002 and 2009) are presented in the Supplementary Figure S1(a–e). For total population and each population group separately, age-specific incidence rates over the three time points showed a general trend of increase with age. As with ASIR, the White population had lower age-specific incidence rates in 2009 compared with 2002.

Mortality

In 1998, there were 1618 BC deaths in SA compared with 2 784 deaths in 2009. Figure 3 presents both BC ASIR and ASMR between 1997 and 2009. Though fairly stable, a concurrent increase in BC ASIR and ASMR was noted between 1997 and 2004. Thereafter line plots suggest stabilisation in both rates. MIR were 0.35 in 1997 for all female BC and increased to a peak of 0.52 in 2007.

Discussion

This study reported on BC incidence and mortality for SA, and explored whether BC patterns differed by ethnicity over the defined time period. There were 85 561 cases of BC reported to the NCR during the study period. For the total population as well as Black, Coloured and Asian groups, ASIR's increased between 1994 and 2009, but decreased for Whites. Black and Coloured populations had the highest increase in LR for the time period. Asians demonstrated a sustained increase of BC as a proportion of all female cancers (1994–2009). White and Asian population groups had higher incidence of BC at an earlier age compared with Blacks

Table 1 BC as a proportion of all female cancers (%)

	All	Black	White	Coloured	Asian
1994	17.3	13.5	19.1	21.9	24.0
1995	15.9	12.8	17.1	12.7	21.4
1996	17.2	14.6	a	a	30.1
1997	16.4	14.2	a	a	28.5
1998	18.6	15.9	19.9	23.9	25.9
1999	19.4	17.2	19.6	24.7	32.9
2000	18.9	16.2	19.4	23.7	31.9
2001	19.1	17.3	18.4	25.5	30.1
2002	20.8	18.9	20.5	25.9	35.0
2003	20.3	18.1	20.4	24.5	35.5
2004	20.0	18.8	18.7	25.9	32.2
2005	20.8	18.7	20.3	27.2	35.7
2006	20.6	18.3	21.2	25.2	32.9
2007	21.0	18.5	21.1	27.1	37.7
2008	20.6	19.0	19.8	25.7	37.6
2009	20.8	19.8	18.9	26.0	39.6

a: Missing data due to the aggregation of White and Coloured groups for NCR reporting in 1996–97.

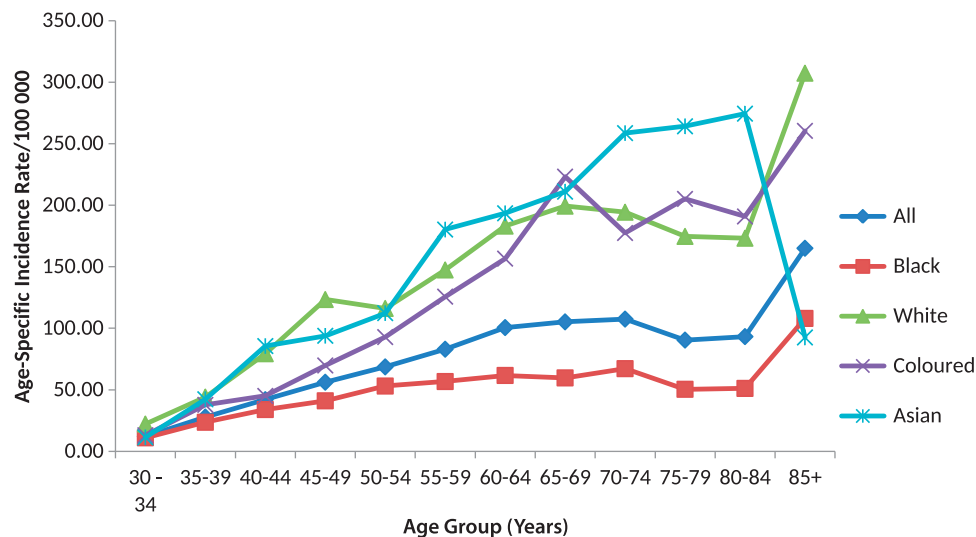


Figure 2 Age-specific incidence rates by ethnic group, 2009

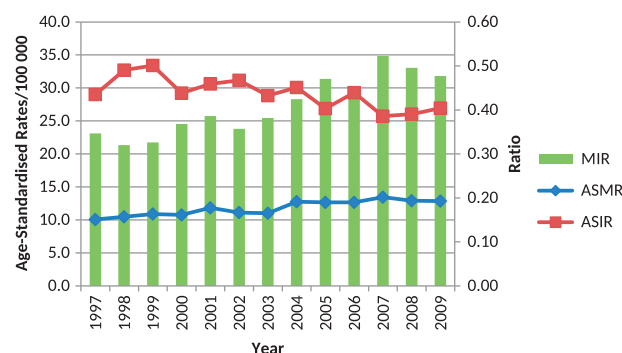


Figure 3 ASIR, ASMR and MIR for all females, 1997–2009

and Coloureds. ASMR for all BC increased between 1994 and 2004 and then stabilised.

Incidences

The ASIR for female BC (all population groups) illustrated a small increase between 1994 and 2009. In 2009, the ASIR for all women was 26.9/100 000. The BC incidence rates reported for SA were comparable to other African countries such as Zimbabwe (28.5/100 000) and Uganda (27.5/100 000) as well as overall for sub-Saharan Africa.¹ This similarity in ASIR from a pathology-based NCR in SA and well-established population-based cancer registries in Zimbabwe and Uganda (both of which have been included in IARC's 'Cancer Incidence in Five Continents') is encouraging, underscoring the validity of SA pathology-based data.

When compared with other MIC's with similar health systems to SA such as Chile (34.8/100 000) and Brazil (59.5/100 000), and Globocan estimates for SA (41.5/100 000),²³ the NCR ASIR were lower. The difference was more striking when BC rates in high-income countries were considered, with the USA and France reporting ASIR's of 92.9/100 000 and 89.7/100 000, respectively.¹ The lower BC ASIR in SA may be a true epidemiological trend but limitations of pathology-based NCR diagnoses, poor reporting from source laboratories and under-diagnosis particularly in rural areas, cannot be ignored. The introduction of population-based cancer surveillance in SA will provide an opportunity to investigate the extent of under-estimation of pathology-based reporting. It is also important to note that Globocan is an estimate based on extrapolation of data for SA.

Interesting differences in BC trends amongst ethnic groups were demonstrated in this analysis. Some of these differences, such as the increase in BC ASIR and LR in non-Whites from 1994, can be attributed to socio-political changes post-democracy (post-1994). Prior to democracy, the SA health system was racially segregated, and health services for the non-White population were systematically underfunded. Access to healthcare facilities specific to BC screening, diagnosis and the treatment for non-White populations was limited. After the first democratic election in 1994, an integrated public healthcare system was introduced with a focus on redressing the inequities of apartheid, resulting in increased availability of and access to healthcare services for the non-White populations.¹² The increases in ASIR between 1994 and 2009 observed for non-Whites could reflect the better diagnosis and reporting into the registry for these previously disadvantaged groups through improved access to healthcare facilities, screening and recording of data. The stabilisation of rates after 1998 could be attributed to the maturation of an equally accessible health system for all population-groups. Though decreases in ASIR for Whites between 1994 and 2009 were observed due to under-reporting from some private laboratories, their rate in 2009 remain significantly higher (more than double) as compared with the Black group but comparable to Coloured and Asian rates.

In this study, the majority of the 'Asian' group consisted of the people of Indian origin. SA has the largest population of people of Indian descent on the African continent, with the initial immigrants having arrived in the 1860s.²⁴ In SA, BC was the commonest female cancer amongst Asian women. Furthermore, the BC proportion was highest in Asians as compared with other race groups between 1994 and 2009. Using the age-specific incidence rates, Asians and Whites showed higher incidences of BC at earlier ages. For Asians this is likely the result of the adoption of westernised lifestyles that have been linked to increased BC risk, such as late age at first pregnancy, decreasing fertility, decreased physical activity, obesity, use of hormonal replacement therapy in post-menopausal women and alcohol consumption.^{25–32} For both groups, where access to private health care is common, early detection, screening and greater patient awareness of BC, may account for the earlier age at diagnosis.

The White population demonstrated an overall decrease in ASIR over the study period. This can be attributed to a decline in reporting from private sector healthcare laboratories from 2004 onwards. Until 2011, when regulations relating to cancer were introduced, reporting to the NCR was on a voluntary basis.³³ As a result, reporting declined from private sector laboratories from 2004, as private laboratories became concerned about the transfer of confidential patient information.¹⁷ It has been shown in SA that a large proportion of those who are able to access private medical care are of White ethnicity. However, this is not consistent with the finding of an increase in ASIR amongst Asians, who would also, in large part access private medical care. The decline in ASIR observed for the White population could, however, also be attributed to possible improvement in modifiable risk factors,^{34,35} although no data could be found on the prevalence of these risk factors specifically for the White population group in SA.

There has been much speculation in SA regarding an increase in BC amongst the Black population attributed to extensive BC awareness campaigns and ad hoc mammography screening by non-governmental organizations. SA currently does not have standardised national clinical BC screening, diagnosis and treatment guidelines approved by the Ministry of Health. Therefore, screening is conducted based on the individual clinical risk and patient request. Whilst an increase in the ASIR in the Black population has been noted, the ASIR has not reached the rates displayed in other population groups. This might be ascribed to under-reporting to the NCR as well as under-diagnosis of cancers in the public healthcare facilities, where most of the Black population seek healthcare.^{17,36}

Mortality

Between 1997 and 2004, there was an increase in BC ASMR which stabilised thereafter to 2009. The increase in mortality after 1997 could be attributed to the increased access to care for all population groups resulting in more BCs being diagnosed and improved cause of death information reported.

The ASMR for SA (12.9/100 000) in 2009 was slightly lower than that of Zimbabwe and Uganda for 2012 (14.0 and 13.6 per 100 000, respectively).¹ It was also lower than the USA (14.9/100 000) and France (16.4/100 000).¹ Only Chile had a lower BC ASMR at 11.5/100 000.¹ Whilst SA's low ASMR may be attributed to good BC outcomes, it is more likely the result of the quality of SA mortality data. IARC has categorised SA mortality data as having 'good coverage' but of 'low quality' based on a report in 2005 that more than 20% of deaths were ill-defined and less than 70% of deaths were accurately recorded.^{1,37} However, it has been demonstrated that neoplasms had the lowest error rate when quality of cause of death certification was evaluated at one SA academic hospital.³⁸

We were unable to observe ethnic differences in mortality as our mortality data were not stratified by population group. However, it

is possible that overall mortality figures mask interesting patterns by ethnicity.

Policy

These data highlight that the BC care and prevention would be improved by an overall national cancer prevention and control policy developed by the Ministry of Health. The lack of BC screening guidelines may contribute to a gap in BC awareness at public healthcare facilities,³⁶ or variance in screening messages delivered by non-governmental organisations (NGO'S) who fill this gap.

The publication of Regulation No. 380 of the National Health Act³³ has significant implications for cancer control in the country. This regulation made cancer a reportable condition and encouraged the implementation of PBCR in SA. This analysis further supports the regulation, by highlighting the need for improved cancer surveillance through the establishment of PBCR's as well as improving the quality of mortality data in SA. Although the NCR has experienced challenges in implementing these regulations,¹⁵ PBCR in SA will generate reliable estimates of cancer incidence through the reporting of all invasive cancers (clinically and pathology diagnosed cancers) in a defined geographic population.

Limitations

The study limitations were the under-estimation of cancers from the pathology-based registration as cancers not biopsied will be missed. In addition, under-reporting occurred from private laboratories from 2004 to 2009 due to concerns regarding reporting of confidential patient information. However, a previous publication has shown that the overall impact of this under-reporting was 4%, whilst a greater impact may be seen in individual ethnic groups (Whites and Asians in particular) depending on their usage of private healthcare facilities.¹⁷ Despite the limitations, this is the only available national data on BC for the country.

Conclusion

This paper demonstrated BC incidence rates are similar to regional rates, but lower than countries with similar health systems. Ethnic differences in BC trends were observed. However, the reasons for observed ethnic differences are unclear.

Supplementary data

Supplementary data are available at *EURPUB* online.

Funding

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Conflict of interest: None declared.

Key points

- Cancer incidence data from Africa is scarce with trend analyses usually covering the USA, European Union, Asia and Latin America. Very little has been reported on South Africa's national cancer burden.
- BC is the commonest female cancer in South Africa with variations in age-standardised incidence rates amongst different ethnic groups.
- Whites and Asians presented higher incidences of BC at an earlier ages compared with Blacks and Coloureds.

- Age-standardised BC mortality rates increased between 1997 and 2004 but stabilised thereafter.
- These data highlight that BC care and prevention would be improved by an overall cancer prevention and control policy, as well as separate BC screening guidelines, developed by the National Department of Health.

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Female lung cancer mortality and long-term exposure to particulate matter in Italy

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Background: Outdoor air pollution and particulate matter (PM) have recently been classified in Group 1 by IARC. In Italy there is no epidemiological study on the association between female lung cancer and PM as measured by the official monitoring stations. **Methods:** We estimated the dose–response relationship between female lung cancer mortality and available long-term outdoor PM10 and/or PM2.5 concentrations for all the Italian province capital city municipalities (respectively, 64 and 32 municipalities). Multiple regression analysis of standardized mortality rates (SMR rates) for the period 2000–11, as a function of PM concentrations, considering percentage of smokers and deprivation index as additional explanatory variables, was performed for PM10 only. **Results:** The number of province capital cities with available PM2.5 data was not sufficient to detect a significant increment of SMR rates as a function of concentrations. An SMR rate increase of 0.325 for 1 µg m⁻³ increment of PM10 concentration was calculated. Moreover, the attributable risk of the overall SMR rates for the two subgroups of municipalities under/equal and above 20 µg m⁻³ value was evaluated. Attributable deaths were computed by both the unitary SMR rate increase and the attributable risk. A rough estimate of the impact of PM10 exposure at level above the WHO guideline value of 20 µg m⁻³ in these 64 municipalities is between 2920 and 3449 lung cancer deaths out of 22 162 (13–16%). **Conclusion:** Maintaining the PM10 concentrations below such WHO recommendation, an overall saving of nearly 300 lung cancer deaths per year in a population of 8 146 520 women living in the municipalities at study has been evaluated.

Introduction

The relevance of the health impact of air pollution is well documented by a number of studies performed in different countries. Chronic exposure to urban air pollution is indeed

responsible for cancer induction, mainly lung cancer, respiratory and cardiovascular diseases and the subsequent burden of deaths.^{1–4} A positive association between air pollution and urinary bladder cancer has also been observed.⁵ Particulate matter (PM) has